BUILD A SOHO NETWORK

Course: Diploma in Cyber Security

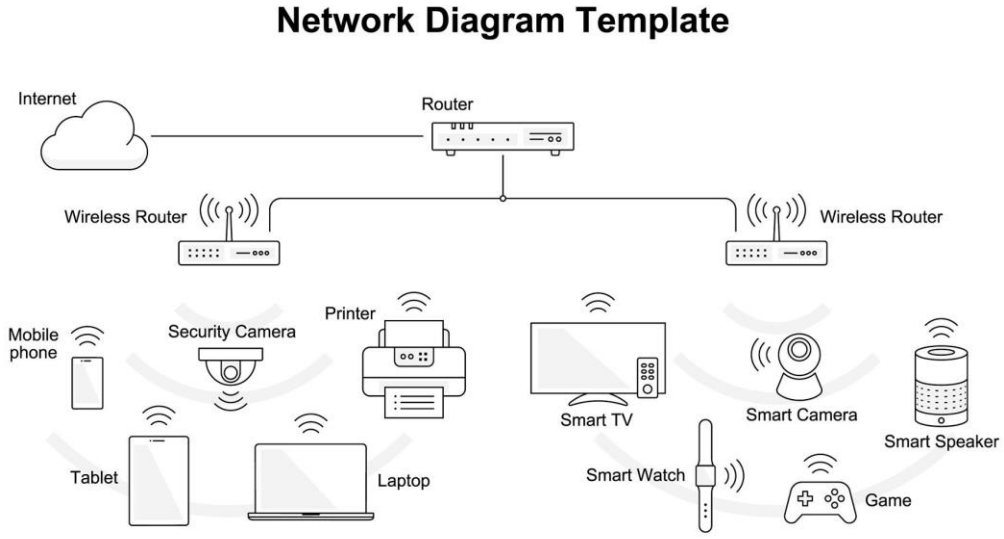
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Introduction

A SOHO network means Small Office/Home Office network. It connects computers, phones, smart TVs and printers so they can share internet and local resources. In this assignment, I planned a small network using a broadband connection, Wi-Fi router and common home devices.

Task 1: Draw the SOHO network diagram

The network contains one broadband internet connection, one Wi-Fi router, two laptops, one smart TV and one printer. The router can connect the laptops, TV and printer by Wi-Fi or Ethernet depending on the device.



NOTE: Internet/Broadband → Wi-Fi Router → Laptops, Smart TV and Printer. Label every connection as Wired or Wireless.]

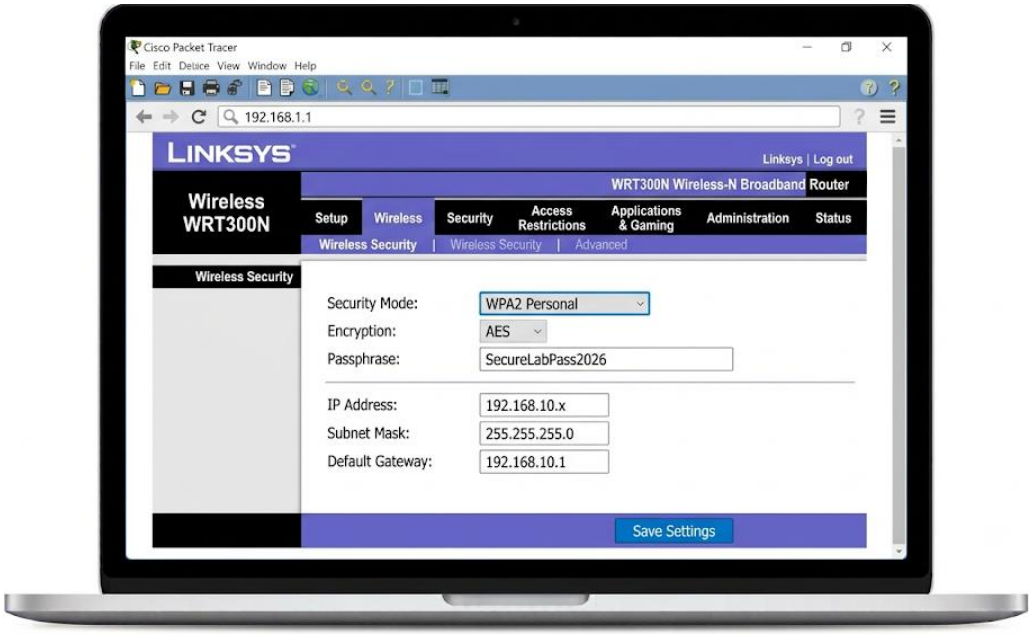
Task 2: List hardware components

Typical hardware needed is a broadband connection/modem if supplied separately, a Wi-Fi router, Ethernet Cat6 cables, and the laptops, smart TV and printer. A small switch can be added if more wired ports are required. A Wi-Fi extender or mesh unit can be used if the coverage area is large.

Task 3: Router configuration

I would open the router settings, change the default admin password, create a unique SSID and select WPA2 security if it is the required lab setting. I would save the

settings and reconnect a test device using the new Wi-Fi name and password.



Task 4: IP address plan

The printer and smart TV can use fixed addresses, while laptops can use DHCP. The MAC addresses below are sample values because the task allows us to make them up.

Task 5: Troubleshooting one laptop

First I would check whether Wi-Fi is switched on and the correct SSID is selected. Then I would check the IP address with ipconfig, ping the router, and finally test internet/DNS access. Possible causes include wrong Wi-Fi password, weak signal, incorrect IP settings, or a router/DNS problem.

Device	Connection	IP Method	Example IP	Example MAC
Laptop 1	Wi-Fi	DHCP	192.168.1.101	00-11-22-33-44-51
Laptop 2	Wi-Fi	DHCP	192.168.1.102	00-11-22-33-44-52
Smart TV	Wi-Fi	Static	192.168.1.20	00-11-22-33-44-53
Printer	Ethernet/Wi-Fi	Static	192.168.1.30	00-11-22-33-44-54

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.